Stephen Stardellis

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EDUCATION

Worcester Polytechnic Institute (WPI) | Worcester, MA | Dean's List | GPA 3.8/4.0 *Graduated March 2026*

Bachelor of Science in Computer Science | Minor in Data Science

EXPERIENCE

Software Engineering Intern | MathWorks Headquarters | Natick, MA *May 2025 - August 2025*

- Migrated 4 on-premises web service applications to AWS cloud infrastructure, increasing scalability and deployment speed.
- Developed and delivered features across a 15-person Agile Scrum team through sprint planning, daily standups, and retrospectives.
- Maintained GitLab CI/CD pipelines to patch CVEs and support secure production releases for 15+ Spring Boot services.
- Automated JUnit-based integration tests across 20+ Spring Boot services and developed Cypress tests to validate frontend behavior, reducing manual regression testing time.
- Validated 7 AWS-based web service releases by verifying deployments through automated tests and monitoring dashboards in Grafana.
- Performed regular smoke tests, including health checks, liveness endpoint validation, and Splunk log verification.

IT Intern | Amphenol Corporation | Wallingford, CT *June 2024 - August 2024*

- Managed 150+ headquarters users across Microsoft Office, Entra, and Admin Center, including licensing and account administration.
- Created anti-phishing campaigns and removed malicious emails using KnowBe4 and Microsoft Defender to educate employees on cybersecurity.
- Developed cost/benefit analysis for Zoom usage at HQ by investigating invoices against other options like Microsoft Teams.
- Onboarded 600+ employees onto Duo authentication to protect SAP users, ensuring a smooth transition through tech support.

PROJECT EXPERIENCE

Mass General Brigham Hospital Kiosk Web Application – PostgreSQL, Express, React, TypeScript, Node.js, AWS, Docker

- Led backend architecture for a 9-person Agile team, designing system structure, producing ERDs, and owning technical documentation.
- Designed a 13-table PostgreSQL schema using Prisma ORM for a hospital kiosk system supporting service requests and pathfinding.
- Built a unified REST API layer in Express.js using a switch-case routing pattern, implementing 15 endpoints to handle form submissions across 6 distinct service request types.
- Implemented admin dashboard features including filtering, inline editing, and CSV import/export, allowing administrators to manage service requests and bulk update hospital map graph data without direct database access.

NYT-Style Web Game "Consensus" – Next.js, Tailwind CSS, Framer Motion, MongoDB, Vercel

- Developed and deployed Consensus, a daily web game where users rank options and compare responses to global averages, attracting 150+ daily active WPI students at peak, built with a 5-person Agile Scrum team.
- Designed the MongoDB schema to support per-puzzle response aggregation, enabling real-time global comparison across all user submissions.
- Built an automated daily archival and scheduling system to rotate puzzles, preserve historical response data, and reset daily state without manual intervention.
- Built an admin dashboard for puzzle creation, content scheduling, and historical response management across the full stack.

Crowd Evacuation Simulator (Major Qualifying Project) – C++, Python, React, TypeScript, NumPy, Matplotlib

- Built a physics-based multi-agent pedestrian evacuation simulator in C++ implementing the Social Force Model (Korhonen et al.), modeling per-agent motive, contact, and frictional forces across a three-ellipse body geometry.
- Designed a multi-source BFS distance field with obstacle inflation and bilinear-interpolated guidance layers for smooth sub-cell agent navigation.
- Created a React/TypeScript scene editor using the HTML5 Canvas API, enabling visual placement and configuration of agents, obstacles, goals, and spawners with export pipelines that generate simulation-ready input files.
- Developed Python visualization tooling using matplotlib and NumPy to render flow field vector maps and replay agent trajectories from output data.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, C/C++, HTML/CSS, SQL, R, MATLAB, Assembly

Frameworks & Tools: React, Next.js, Node.js, Express, Spring Boot, Prisma ORM, Tailwind CSS, Framer Motion, Shadcn UI, HTML5 Canvas, PostgreSQL, MongoDB, Docker, AWS (EC2, RDS), GitLab CI/CD, GitHub/GitLab, Cypress, JUnit, GDB Debugger, Ubuntu, Postman, Grafana, Splunk, Jira, Confluence, TeamCity, NumPy, Matplotlib, Gemini API, Linux Kernel

Relevant Coursework: Webware, Software Engineering, Database Systems I & II, Systems Programming, Analysis of Algorithms, Operating Systems, Machine Learning, Data Science I & II, Machine Organization & Assembly Language